3 Points Each

2003

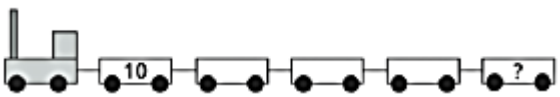
1. The picture shows the letter U drawn on grid paper. How many squares does the letter U cover?

(A) 10
(B) 8
(C) 11
(D) 13
(E) 12

2. What is the result of
0 + 1 + 2 + 3 + 4 – 3 – 2 – 1 – 0?

(A) 0
(B) 2
(C) 4
(D) 10
(E) 16

3. The first car of the train, right behind the engine, contains 10 boxes. In each of the following cars there are twice as many boxes as in the car in front of it. How many boxes are there in the fifth car?



(A) 100
(B) 120
(C) 140
(D) 160
(E) 180

4. Sophia is drawing kangaroos. The first one is blue, the next one green, the one after it red, the fourth one yellow, and then again blue, green, red, yellow, and so on, in the same order. What color will the seventeenth kangaroo be?

(A) blue
(B) green
(C) red
(D) black
(E) Yellow

5. In the teachers' lounge there are 6 tables with 4 chairs each, 4 tables with 2 chairs each, and 3 tables with 6 chairs each. How many chairs are there in the lounge?

(A) 40
(B) 25
(C) 50
(D) 36
(E) 44

6. In one of these pictures, there are three times as many hearts as other shapes. Which picture is it?

(A) 

(B) 

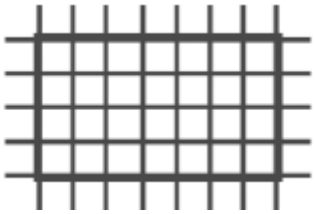
(C) 

(D) 

(E) 

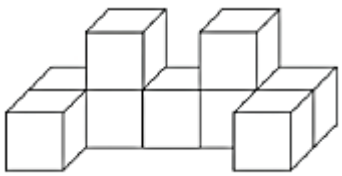
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7. A rectangle with dimensions 7 x 4 was outlined on grid paper. How many squares will a diagonal of this rectangle intersect?



(A) 8
(B) 9
(C) 10
(D) 11
(E) 12

8. The figure presented in the picture, which is made out of a certain number of identical cubes, weighs 189 grams. How much does one cube weigh?



(A) 29 grams
(B) 25 grams
(C) 21 grams
(D) 19 grams
(E) 17 grams

4 Points Each

9. Philip wrote down consecutive natural numbers starting with 3 until he had written 35 digits. What was the greatest number that Philip wrote down?

(A) 12
(B) 22
(C) 23
(D) 28
(E) 35

10. Anna fell asleep at 9:30 p.m. and woke up at 6:45 a.m. the next day. Her little brother Peter slept 1 hour and 50 minutes longer. How long did Peter sleep?

(A) 30 hr 5 min
(B) 11 hr 35 min
(C) 11 hr 5 min
(D) 9 hr 5 min
(E) 8 hr 35 min

11. A pattern, the beginning and the end of which is shown in the picture, is made up of alternating black and white bars. There are 17 bars altogether. The bars on both ends are black. How many white bars are there in the pattern?



(A) 9
(B) 16
(C) 7
(D) 8
(E) 15

12. Jumping Kangaroo is practicing for the animal Olympics. His longest jump during the training was 55 dm 50 mm long, but in the finals of the Olympics he won with a jump that was 123 cm longer. How long was Jumping Kangaroo's longest jump during the Olympics?

(Remember that 1 m = 10 dm, 1 dm = 10 cm, and 1 cm = 10 mm)

(A) 6 m 78 cm
(B) 5 m 73 cm
(C) 5 m 55 cm
(D) 11 m 28 cm
(E) 7 m 23 cm

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13. Paul chose a certain number, subtracted 203 from it, and then added 2003 to the difference. His final result was 20003. What number did Paul choose at the beginning?

(A) 23
(B) 17797
(C) 18203
(D) 21803
(E) 22209

14. Barbara's electronic watch shows time in 24-hour format. She likes to add the digits showing the current time. For example, when the watch shows 21:17, she gets the number 11 as the result. What is the greatest sum she can get?

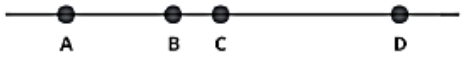
(Hint: In some countries and sometimes in the US, instead of saying "It is 1 p.m.," people may say, "It is 13:00." When it is 2 p.m., they may say, "It is 14:00," and when it is 12 a.m., they may say, "It is 24:00." In this problem, 21:17 means 9:17 p.m. Time expressed in this way is sometimes called "military time.")

(A) 24
(B) 36
(C) 19
(D) 25
(E) 23

15. Mark said to his friends, "If I had picked twice as many apples as I did, I would have 24 more apples than I have now." How many apples did Mark pick?

(A) 48
(B) 24
(C) 42
(D) 12
(E) 36

16. Points A, B, C, and D, all of which lie on a straight line, are marked in the picture be- low. The distance between points A and C is 10 m, between B and D is 15 m, and between A and D is 22 m, What is the distance between points B and C?



(A) 1 m
(B) 2 m
(C) 3 m
(D) 4 m
(E) 5 m

5 Points Each

17. There are 29 students in a certain class. 12 of the students have a sister and 18 of the students have a brother. In this class, only Tania, Barbara, and Anna do not have any siblings. How many students from this class have both a brother and a sister?

(A) none
(B) 1
(C) 3
(D) 4
(E) 6

18. Daniel has 11 pieces of paper. He cut some of them into three parts and now he has 29 pieces of paper. How many pieces of paper did he cut?

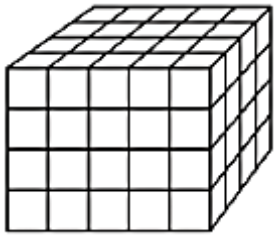
(A) 3
(B) 2
(C) 8
(D) 11
(E) 9

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19. John bought 3 kinds of cookies: large, medium, and small. The large cookies cost 4 dollars each, the medium 2 dollars each, and the small 1 dollar each. Altogether, John bought 10 cookies and paid 16 dollars. How many large cookies did he buy?

(A) 5
(B) 4
(C) 3
(D) 2
(E) 1

20. Christopher built the rectangular prism shown in the picture using red and blue cubes of identical size. The outer walls of this prism are red but all the inner cubes are blue. How many blue cubes did Christopher use in this construction?

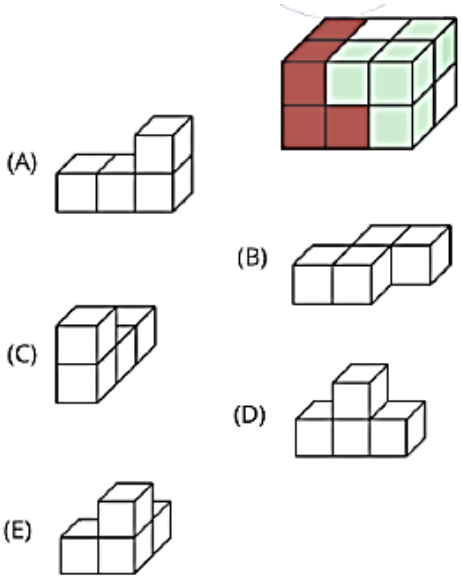


(A) 12
(B) 16
(C) 22
(D) 26
(E) 32

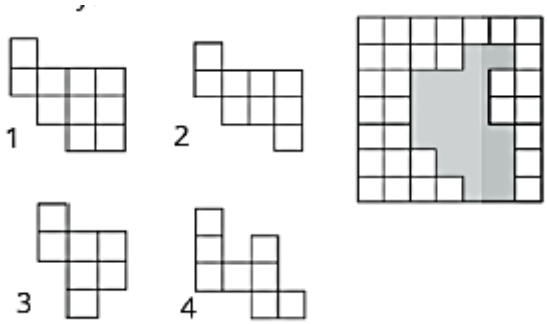
21. Jerry is planning to buy some basketballs. If he buys 5 basketballs, he will have 10 dollars left over, and if he buys 7 basketballs, he will have to borrow 22 dollars. How many dollars does one basketball cost?

(A) 11
(B) 16
(C) 22
(D) 26
(E) 32

22. Mark built a rectangular prism using 3 blocks, each of which is made out of 4 small cubes connected in various ways. The two colored blocks can be fully seen in the picture. Which is the third (white) block with only two visible sides?



23. Two pieces that made up the shaded region were cut out from a square puzzle (see the picture). Which two of the pieces are they?



(A) 1 and 3
(B) 2 and 4
(C) 2 and 3
(D) 1 and 4
(E) 3 and 4

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24. At the toy store, among other things, you can buy dogs, bears, and kangaroos. Three dogs and two bears together cost as much as four kangaroos. For the same amount of money, you can buy one dog and three bears. Which of the statements below is true?

(A) A dog is twice as expensive as a bear.
(B) A bear is twice as expensive as a dog.
(C) The prices of a dog and of a bear are identical.
(D) A bear is three times as expensive as a dog.
(E) A dog is three times as expensive as a bear.

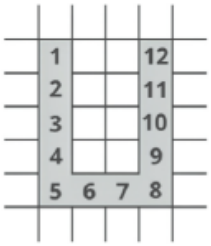
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Answers key:

1. E
2. C
3. D
4. A
5. C
6. D
7. C
8. C
9. C
10. C
11. D
12. A
13. C
14. A
15. B
16. C
17. D
18. E
19. E
20. A
21. B
22. D
23. A
24. B

3 Point Solutions

1 (E) 12



Count the squares of the shaded part by using the grid to help.

2 (C) 4

To find the result of 0 + 1 + 2 + 3 + 4 – 3 – 2 – 1 – 0 quickly, notice that all the numbers that are added except for 4 are also subtracted.

3 (D) 160

Since each car has twice as many boxes as the car in front of it, multiply the number in each car by 2 to get the number of boxes in the next car.



4 (A) blue

blue green red yellow
1 2 3 4

\- 5 6 7 8 - 9 10 11 12 - 13 14 15 16 - 17

The pattern consists of four colors repeated in the same order. Because 17 is one more than 16, which is a multiple of 4, we can determine that the 17th color is the same as the first color in the sequence, which is blue.

5 (C) 50

Multiply the number of tables with the same number of chairs by the number of chairs next to the tables. 6 tables with 4 chairs each gives us 6 × 4 = 24 chairs. 4 tables with 2 chairs each gives us 4 × 2 = 8 chairs. 3 tables with 6 chairs each gives us 3 × 6 = 18 chairs. The total number of chairs in the teachers' lounge is 24 + 8 + 18 = 50 chairs.

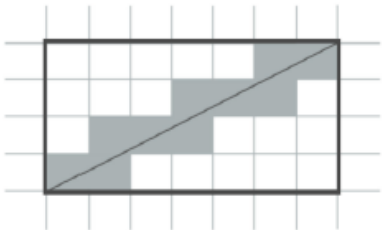
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6 (D)



For every shape that is not a heart there have to be 3 hearts in order for the picture to have 3 times as many hearts as other shapes. Picture D has 9 hearts and 3 other shapes. 3 × 3 = 9.

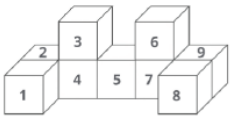
7 (C) 10



Draw a diagonal as shown and count the squares it passes through. The diagonal passes through 10 squares of the grid.

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8 **(C) 21 grams**
There are 9 cubes. Since they are identical, each cube weighs the same amount. The weight of all the cubes is 189 grams or 189 g. The total weight divided by the number of cubes will equal the weight of one cube. 189 g divided by 9 equals to 21 g.



$$\begin{array}{r} 21 \\ 9\overline{)189} \\ -18 \\ \hline 09 \\ -\,9 \\ \hline 0 \end{array}$$

4 Point Solutions

9 **(C) 23**
Consecutive numbers are numbers in order such as 1, 2, 3, etc. Natural numbers are the normal counting numbers. Philip wrote down seven single-digit numbers from 3 to 9. He still needed to write 35 – 7 = 28 more digits, which means that he wrote fourteen more two-digit numbers, from 10 to 23, so 23 is the greatest number that Philip wrote down.

10 **(C) 11 hr 5 min**



Anna slept from 9:30 p.m. to 6:45 a.m. as indicated by the shaded part of the clock, so she slept 9 hours and 15 minutes. Since Peter slept 1 hour and 50 minutes longer than Anna, he slept for 11 hours and 5 minutes. Remember that 15 minutes + 50 minutes is 65 minutes, which is 1 hour and 5 minutes.

11 **(D) 8**
We can describe the pattern as 8 pairs of black-white bars plus the last black bar, so there are 8 white bars in the pattern.

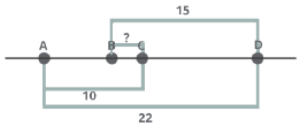
12 **(A) 6 m 78 cm**
55 dm = 55 × 10 cm = 550 cm and 50 mm = 5 × 10 mm = 5 cm, so Jumping Kangaroo's longest jump during the Olympics was (550 cm + 5 cm) + 123 cm = 678 cm, which is 6 m 78 cm.

13 **(C) 18203**
Perform the operations in reverse order. From the final result of 20003 subtract 2003, which equals 18000. To get the number which Paul chose at the beginning, add 203 to 18000. The result is 18203.

14 **(A) 24**
For the hour part, we need to find the greatest sum of the digits on the interval from 00 to 24. For the hours from 20 to 24, the largest sum of digits is 2 + 4 = 6. For the hours from 00 to 19 the largest sum is 1 + 9 = 10, so 10 is the largest sum for the hour part. For the minute part, we need to find the greatest sum of the digits on the interval from 00 to 59, which happens to be 59, with 5 + 9 =14. Barbara will get the greatest sum of the digits on her electronic watch at 19:59 (which is 7:59 p.m.), and that sum is 1 + 9 + 5 + 9 = 24.

15 **(B) 24**
24 is the number that would double the current number of apples. So, the number of apples Mark actually did pick is also 24.

16 **(C) 3 m**



Since the distance from A to D is 22 m, and the distance from A to C is 10 m, then the distance from C to D is 22 m – 10 m = 12 m. Since the distance from B to D is 15 m, and the distance from C to D is 12 m, then the distance from B to C is 15 m – 12 m = 3 m.

5 Point Solutions

17 **(D) 4**
Three students do not have siblings, therefore there are 29 – 3 = 26 students who have siblings. The number of sisters and brothers the students have is 12 + 18 = 30 siblings. Because there are more siblings than students, there will be some students who have both a sister and a brother. There are 30 – 26 = 4 students with both a sister and a brother.

18 **(E) 9**
Once Daniel cuts a piece of paper into three parts, instead of having one piece he now has three. That is two more pieces than he started with, even though they are now smaller.
If we cut 1 piece of paper, we have 11 + 2 pieces.
If we cut 2 pieces of paper, we have 11 + 2 + 2 pieces.
If we cut 3 pieces of paper, we have 11 + 2 + 2 + 2 pieces, and so on.

Thus, 11 + 2 × the number of all pieces cut is 29. Hence, the number of all pieces cut is half of 29 − 11 = 18, which is 9.

19 **(E) 1**
John bought 3 kinds of cookies: large, medium, and small, so he had to buy at least one of each and pay 4 + 2 + 1 = 7 dollars for these 3 cookies. After that he had 9 dollars left to buy 7 more cookies. 6 cookies of any kind cost at least 6 dollars, so the seventh cookie cannot be a large one since 6 dollars + 4 dollars is 10 dollars (only 9 dollars are left). Therefore, John bought only 1 large cookie. (If we solve the problem further, we can find that he bought 1 large cookie, 3 medium cookies, and 6 small cookies.)

20 **(A) 12**
Christopher built a rectangular prism with the dimensions of 5 × 4 × 4. To find the number of all the blue inner cubes, we need to remove all the red outer walls (or layers): front and back, top and bottom, left and right sides. Basically, we are reducing the dimensions of Christopher's rectangular prism by 2 units for each dimension. The resulting dimensions are 3 × 2 × 2, so the number of blue inner cubes is 12.

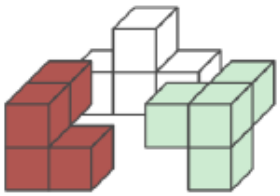
21 **(B) 16**
Jerry's purchase plans differ by 2 basketballs. To buy those two additional basketballs he needs to borrow 22 dollars and also use the 10 dollars he would have left over had he only bought 5 basketballs. The cost of buying the two extra basketballs is 22 + 10 = 32 dollars. Therefore, one basketball costs 16 dollars.

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22 (D)



Notice that the bottom of the back wall is made of 3 white cubes and another white cube is visible above its center.



23 (A) 1 and 3



Draw in the lines of the grid over the shaded region. Piece 1 is the only one that can fill in the bottom of the shaded region; it is marked in blue here. Piece 3 needs to be rotated clockwise to fill in the remainder of the shaded region; it is marked in pink.

24 (B) A bear is twice as expensive as a dog. 3 dogs and 2 bears together cost as much as 1 dog and 3 bears since each collection costs as much as 4 kangaroos. Removing 1 dog and 2 bears from each collection results in 2 dogs being priced as much as 1 bear, so a bear is twice as expensive as a dog.

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